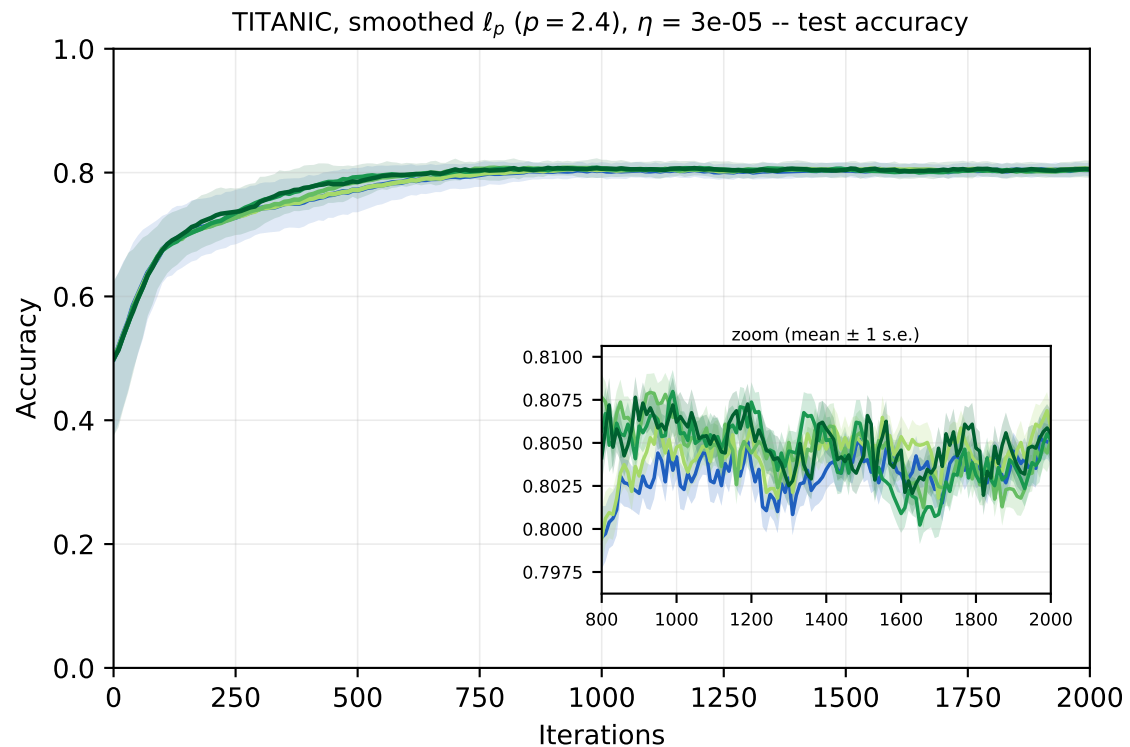
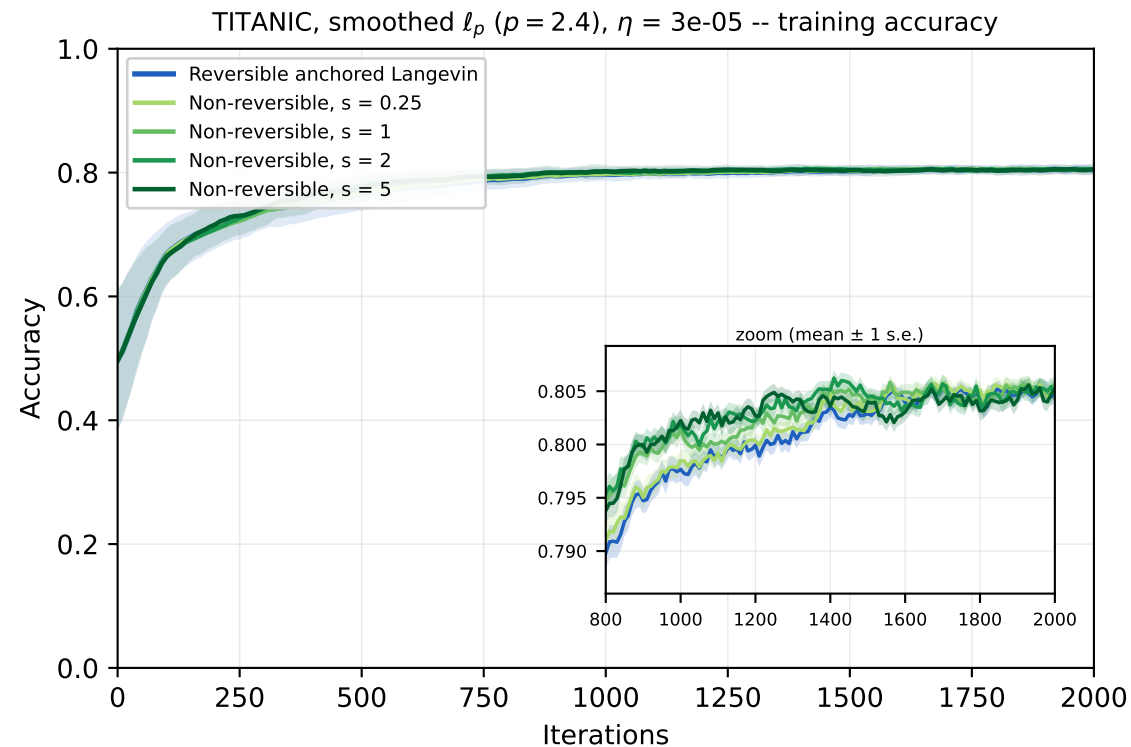


titanic_lp: accuracy vs block strength at $\eta = 3\text{e-}05$



TITANIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{(p/2)} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept $w_0 + 9$ features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \text{sigma}^2) + \text{lambda} \sum_{|j| \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \text{delta}^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\text{sigma} = 10$, $\text{lambda}_{\text{lasso}} = 7.12 = 0.01 n_{\text{train}}$, $\text{delta} = 0.0108$ (a in $[0.50, 1]$). Step $\text{eta} = 3\text{e-}05$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Bands: mean ± 1 SD (ddof = 1) for the reversible arm and for $s = 5$ -- repeat-run variability at a fixed split, not a confidence or credible interval. The whole $\text{eta} \times s$ grid is reported in `results/grid/grid.csv`; nothing here was selected.